Simrad HALO-6 Pulse Compression Radar

Technical Maintenance and Service Log

Equipment and Service Information

Equipment Model:	Simrad HALO-6 Pulse Compression Radar (w/ 6-foot open array)
Vessel Name / ID:	Brother Printer-172
Scanner Serial Number:	5
RI-12 Interface Box S/N:	N/A
Service Date:	2026-02-23
Technician Name:	Kelvin
Technician Signature:	____________________________

Maintenance Work Performed

Change radar rotation motor

Parts Replaced:
KKWS-102 Bearing

Hours Spent: 3 hours

Additional Comments:
General grease replace

Service Status:	**Pass**

Preventive Maintenance Checklist

Perform these checks periodically to ensure optimal performance and longevity of the radar system. Status options: Pass, Fail, N/A (Not Applicable).

Task #	Category	Task Description	Status	Comments
A.1	A. Physical & Mechanical Inspection	Visually inspect the 6-foot open array antenna for cracks, physical damage, or delamination. Ensure it is clean and free of debris.		
A.2		Clean the antenna array using a soft cloth, fresh water, and a mild, non-abrasive soap. Rinse thoroughly. Do not use chemical solvents or high-pressure washe		
A.3		Inspect the pedestal housing for cracks, seal integrity, and signs of water ingress. Check that all covers are secure.		
A.4		Check all mounting bolts for the pedestal for proper tightness and signs of corrosion. Ensure marine-grade stainless steel hardware is used.		
A.5		With the system powered down, manually check that the antenna can swing freely without obstruction or excessive resistance.		
B.1	B. Electrical & Network Inspection	Inspect the main power cable to the RI-12 Interface Box. Check for chafing, cuts, UV damage, and corrosion at the connection points. Ensure connections are t		
B.2		Inspect the cable between the RI-12 Interface Box and the scanner pedestal. Check for damage to the cable and connectors (RJ45). Ensure a secure connectio		
B.3		Inspect the Ethernet cable connecting the RI-12 to the MFD or network switch. Check for secure connections and cable integrity.		
B.4		Open the RI-12 Interface Box. Visually inspect internal connections for tightness, signs of corrosion, or moisture. Ensure the lid seal is in good condition.		
B.5		With the system on, use a multimeter to measure the DC voltage at the RI-12 power input terminals. Verify it is within the specified range (typically 12/24V DC).		
B.6		If connected, inspect the NMEA 2000 backbone. Verify the presence of exactly two terminators, secure T-connectors, and a stable power supply (9-16V DC).		
C.1	C. System & Software Checks	Check the radar's installed software version via the MFD's network device list. Compare with the latest version available on the Simrad Yachting support websit		
C.2		Power on the MFD and radar system. Confirm the radar is detected by the MFD within the normal 30-40 second startup time and does not display a "No Scann		
C.3		Place the radar in Transmit (Tx) mode. Confirm the open array antenna begins to rotate smoothly and quietly. Listen for any unusual noises from the pedestal m		
C.4		Verify a clear radar picture is painted on the MFD. Check for good target definition at both short (e.g., <1 NM) and long ranges (e.g., >12 NM).		
C.5		Briefly test key operational modes such as Harbour, Offshore, Weather, and Bird to confirm they activate and adjust the display as expected.		
C.6		Activate Simultaneous Dual Range operation. Confirm the MFD displays two independent radar ranges and that both update correctly.		

Corrective Maintenance Log

Reported Fault / Symptom Description:

Change radar rotation motor

Parts Replaced:
KKWS-102 Bearing

Hours Spent: 3 hours

Additional Comments:
General grease replace

Troubleshooting Steps & Findings

Follow these guided steps based on the reported symptom. Document all actions and findings.

Step	Action Taken / Check Performed	Result / Finding	Corrective Action / Notes
1.1 Check dedicated breaker/fuse for marine electronics.			
1.2 Confirm voltage at the RI-12 Interface Box power input.		Voltage Reading: _____ V	
1.3 Inspect power cable from source to RI-12 for damage.			
1.4 Power cycle the entire system. Wait 60 seconds.			
2.1 Check MFD > Settings > Network > Device List.			
2.2 Inspect Ethernet cable from RI-12 to MFD/switch. Reseat ends.			
2.3 Inspect interconnect cable from RI-12 to scanner pedestal.		Common point of failure.	
3.1 Confirm radar is in Transmit (Tx) mode on MFD.			
3.2 Observe scanner pedestal - confirm antenna rotating.			
4.1 Listen for motor drive sound in pedestal when in Tx mode.			
4.2 Power down - inspect antenna path for physical obstructions.			

Final Diagnosis & Resolution Summary:

Final System Test & Verification

To be completed after any maintenance or corrective action.

Test #	Verification Check	Pass / Fail	Comments
1	System powers on correctly and radar is detected by MFD without errors.		
2	Radar transmits and antenna rotates smoothly.		
3	Clear radar image is displayed with good target separation.		
4	All operational modes (Harbour, Offshore, etc.) and Dual Range function correctly.		
5	System operates for a minimum of 15 minutes without recurrence of the original fault.		

Technician:

Signature: ____________________________
Name: Kelvin
Date: 2026-02-23

Customer / Supervisor:

Signature: ____________________________
Name: ____________________________
Date: ____________________________